

H524 Introduction to Biostatistics
Week 6: Power, Two-Sample Tests, and ANOVA
Homework Assignment
Fall 2025

Due Date: End of Week 6

Total Points: 28 points

Format: Submit through Canvas

Time Limit: 60 minutes

Instructions

- Show all work for full credit on numerical questions
- Round numerical answers to 2 decimal places unless otherwise specified
- Include R code where specified
- You may use R to perform calculations
- For multiple choice, select the BEST answer

Multiple Choice Questions (10 points total, 2 points each)

Question 1: Statistical power is defined as:

- A. The probability of making a Type I error
- B. The probability of correctly rejecting a false null hypothesis
- C. The probability of failing to reject a true null hypothesis
- D. The significance level of the test

Question 2: Which of the following will *increase* the statistical power of a study?

- A. Decreasing the sample size
- B. Increasing the population variance
- C. Increasing the significance level (alpha)
- D. Decreasing the effect size

Question 3: A researcher measures blood pressure in 15 patients before and after a medication intervention. Which test should be used?

- A. One-sample t-test

- B. Independent two-sample t-test
- C. Paired t-test
- D. One-way ANOVA

Question 4: When comparing means across 4 independent groups, why is ANOVA preferred over multiple pairwise t-tests?

- A. ANOVA is easier to calculate
- B. ANOVA requires smaller sample sizes
- C. ANOVA controls the family-wise Type I error rate
- D. ANOVA doesn't require assumptions

Question 5: An ANOVA test comparing 5 groups yields $F = 3.45$ with $p = 0.02$. What can you conclude?

- A. All five group means are significantly different from each other
- B. At least one group mean differs from the others
- C. The first group differs significantly from the last group
- D. Groups 1 and 2 have equal means

Numerical Questions (15 points total)

Question 6: Sample Size for Power (3 points)

A researcher is planning a study to test whether a new medication reduces systolic blood pressure. Based on pilot data, the population standard deviation is estimated to be $\sigma = 20$ mmHg. The researcher wants to detect a mean reduction of 10 mmHg from the population mean of 145 mmHg.

Using a significance level of $\alpha = 0.05$ (two-sided) and desired power of 0.80, calculate the required sample size using R's `power.t.test()` function for a one-sample t-test.

Show your R code:

What is the required sample size (round UP to the nearest whole number)?

Answer: _____ subjects

Question 7: Two-Sample t-Test and ANOVA (3 points)

A researcher compares the effectiveness of three different exercise programs on weight loss (in kg) over 8 weeks. The data are:

Program A	Program B	Program C
4.2	6.1	7.8
3.8	5.8	8.2
4.5	6.4	7.5
4.1	5.9	8.0
4.3	6.2	7.9

Part A (1.5 points): Perform a one-way ANOVA to test if there is a significant difference in mean weight loss across the three programs. Use $\alpha = 0.05$.

Show your R code to create the data and perform ANOVA:

What is the F-statistic (round to 2 decimal places)?

Answer: F = _____

Part B (1.5 points): Compare ONLY Program A vs. Program C using a two-sample t-test (assume equal variances, two-sided test).

Show your R code:

What is the p-value (round to 4 decimal places)?

Answer: p-value = _____

Question 8: Pooled Standard Deviation (3 points)

Two independent samples are collected to compare cholesterol levels between two diets:

- Diet 1: $n_1 = 12$, $s_1 = 18.5$ mg/dL
- Diet 2: $n_2 = 15$, $s_2 = 21.3$ mg/dL

Calculate the pooled standard deviation (s_p) using the formula:

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

Show your calculation (R code or by hand):

Answer: $s_p =$ _____ mg/dL (round to 2 decimal places)

Question 9: Test Statistic Calculation (3 points)

A researcher compares mean test scores between two teaching methods. The summary statistics are:

- Method 1: $n_1 = 20$, $\bar{x}_1 = 85.3$, $s_1 = 12.4$
- Method 2: $n_2 = 18$, $\bar{x}_2 = 78.6$, $s_2 = 14.1$

Calculate the test statistic (t-value) for a two-sample t-test (assume equal variances) using the formula:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

where $s_p = \sqrt{\frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}}$

Show your calculation (R code or by hand):

Answer: $t =$ _____ (round to 2 decimal places)

Question 10: Confidence Interval for Mean Difference (3 points)

Using the data from Question 9 (Teaching Methods), construct a 95% confidence interval for the mean difference in test scores ($\mu_1 - \mu_2$).

Use the formula:

$$CI = (\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2, df} \times SE$$

where $SE = s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$ and $df = n_1 + n_2 - 2$

Or use R to calculate (show your work either way):

Answer: 95% CI = (_____, _____) (round to 2 decimal places)

Interpretation: Based on this confidence interval, is there evidence of a significant difference between the two teaching methods at $\alpha = 0.05$? Explain briefly.

Question 11: Power Calculation (3 points)

A researcher is planning a one-sample study to test whether a new intervention changes blood glucose levels. Based on prior research:

- Expected sample size: $n = 30$
- Expected mean difference to detect: $\delta = 12$ mg/dL
- Population standard deviation: $\sigma = 18$ mg/dL
- Significance level: $\alpha = 0.05$ (two-sided)

Calculate the statistical power of this study using R's `power.t.test()` function.

Show your R code:

Answer: Power = _____ (round to 2 decimal places)

Note: Express your answer as a proportion (e.g., 0.85), not as a percentage.